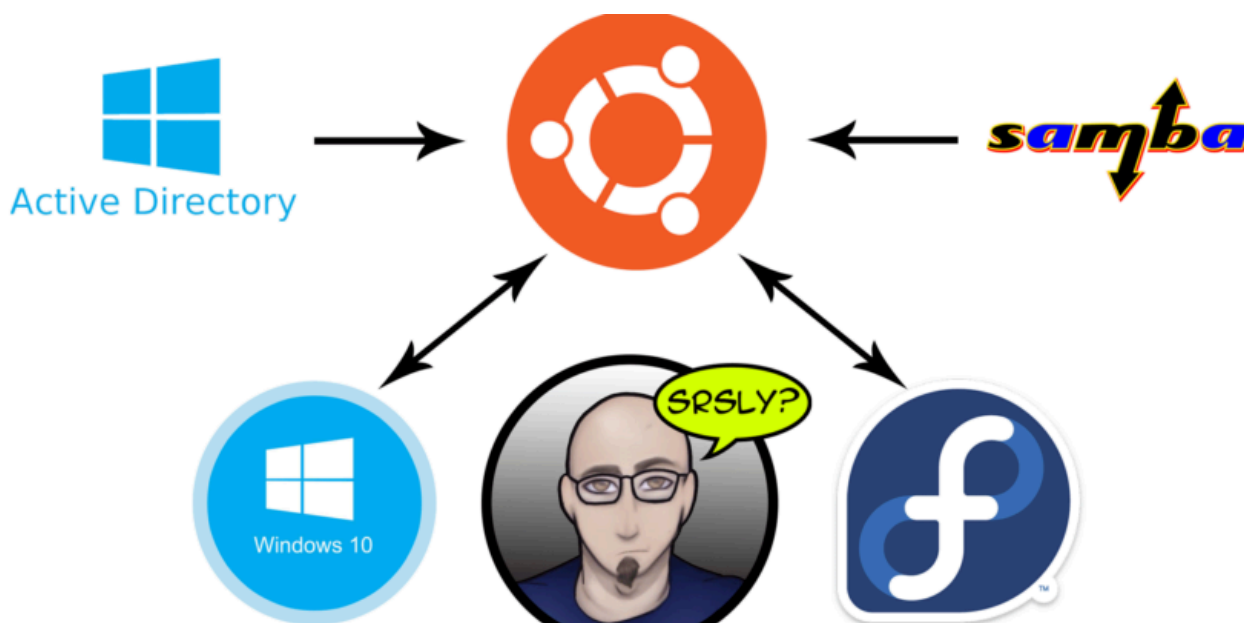


Samba Based Active Directory on Ubuntu 22.04

 considerednormal.com/2022/11/samba-based-active-directory-on-ubuntu-22-04

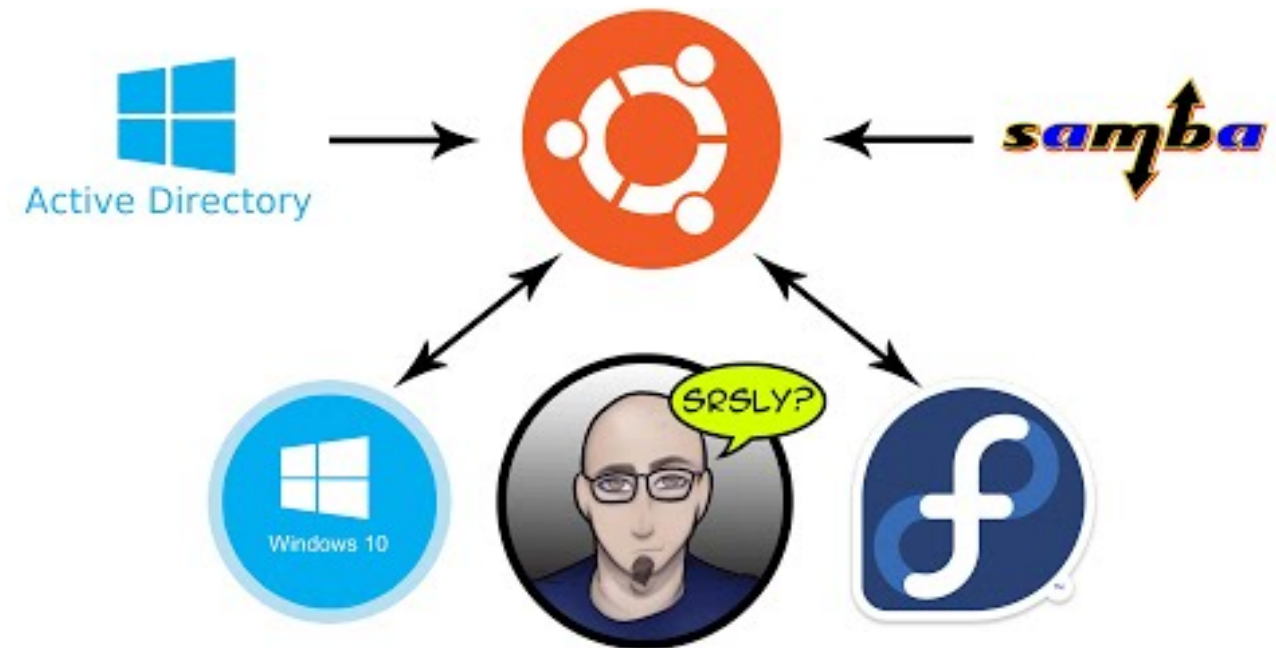
8 ноября 2022 г.



Preface

I have been waiting for a long time to write this post. In order to test this out, I needed to upgrade my home lab. I had a VM machine running [XCP-NG](#) (I tried Proxmox, it's just not the same), but it had an old dual core Intel and 8GB of ram, so not much I could do with other than host a good single VM... which is pointless, the server might as well be the function of that single VM. So recently I was able to upgrade my main system to a Ryzen 5 5600X with a new mobo, RAM, and PSU, so I figured I would use my old Ryzen 5 1600X in the server. So now I have 6 cores which translated to 12 Threads (each of which can be used as a virtual CPU), 32GB of RAM, and more then enough resources to make a few VMs with multiple purposes. The first of which of course was a Linux based Active Directory.

Video



[Watch on YouTube](#)

Please note, the following commands are meant to accompany the video, some context may be missing.

Prerequisites

- Ubuntu Server (20.04/22.04)
- A user account with **sudo** privileges
- A Windows 10 PRO computer on the same network
- A Linux Desktop on the same server (Fedora or Ubuntu based)

In this example will be using Ubuntu 22.04 Server for the Active Directory. We will be connecting to it with a Windows 10 PRO client as well as Fedora as the Linux based client

DC Server Setup

Set the Server Hostname

For this demonstration we will be using the hostname dc1 for the Ubuntu server

- 1 # set up the hostname
- 2 hostnamectl set-hostname dc1

Add Server to Hosts File

We will also need to modify the hosts file, add the following line to /etc/hosts

- 1 # setup FQDN dc1.cn.lan
- 2 192.168.0.254 dc1.cn.lan dc1

Verify Hostname

You can quickly verify everything worked with the following command

- 1 # verify FQDN
- 2 hostname -f
- 3 # verify FQDN is resolved to the Samba IP address
- 4 ping -c3 dc1.cn.lan
- 5

Disable the DNS Resolver

Disable the DNS Resolver and unlink the config

- 1 # stop and disable systemd-resolved service
- 2 sudo systemctl disable --now systemd-resolved
- 3 # remove the symlink file /etc/resolv.conf
- 4 sudo unlink /etc/resolv.conf
- 5

Create our own Resolv.conf

- 1 # create a new /etc/resolv.conf file
- 2 touch /etc/resolv.conf

Enter the following into /etc/resolv.conf

```
1 # Samba server IP address
2 nameserver 192.168.0.254
3 # fallback resolver
4 nameserver 9.9.9.9
5 # main domain for Samba
6 search cn.lan
7
8
```

Make the file immutable

As we do not want anything to change this file automatically, we make it immutable so that it does not get modified

```
1 # add attribute immutable to the file /etc/resolv.conf
2 sudo chattr +i /etc/resolv.conf
```

Installing Samba

```
1 sudo apt update
2 sudo apt install -y acl attr samba samba-dsdb-modules samba-vfs-modules
   smbclient winbind libpam-winbind libnss-winbind libpam-krb5 krb5-config krb5-user
   dnsutils chrony net-tools
```

Enter the appropriate info

Disable Samba Services

```
1 # stop and disable samba services - smbd, nmbd, and winbind
2 sudo systemctl disable --now smbd nmbd winbind
```

Activate samba-ad-dc

```
1 # activate samba-ad-dc service
2 sudo systemctl unmask samba-ad-dc
3 # enable samba-ad-dc service
4 sudo systemctl enable samba-ad-dc
5
```

Configuring Samba Active Directory

First backup the original smb.conf file

- 1 `sudo mv /etc/samba/smb.conf /etc/samba/smb.conf.bak`

Provision the Active Directory

- 1 `# provisioning Samba Active Directory`
- 2 `sudo samba-tool domain provision`

Enter the following info

Backup and replace the Kerberos Config

Start samba-ad-dc

- 1 `# start samba-ad-dc service`
- 2 `sudo systemctl start samba-ad-dc`
- 3 `# verify samba-ad-dc service`
- 4 `sudo systemctl status samba-ad-dc`
- 5

Set up Time Synchronization

Set permissions

- 1 `# allow group _chrony to read the directory ntp_signd`
- 2 `sudo chown root:_chrony /var/lib/samba/ntp_signd/`
- 3 `# change the permission of the directory ntp_signd`
- 4 `sudo chmod 750 /var/lib/samba/ntp_signd/`
- 5

Update Chrony

add the following to `/etc/chrony/chrony.conf`

Verifying Samba Active Directory

Run the following to verify

```
1 # verify domain example.lan
2 host -t A cn.lan
3 # verify domain dc1.example.lan
4 host -t A dc1.cn.lan
5
```

Then verify the Kerberos and ldap services

```
1 # verify SRV record for _kerberos
2 host -t SRV _kerberos._udp.cn.lan
3 # verify SRV record for _ldap
4 host -t SRV _ldap._tcp.cn.lan
5
```

Then verify the Samba resources

```
1 # verify SRV record for _kerberos
2 host -t SRV _kerberos._udp.cn.lan
3 # verify SRV record for _ldap
4 host -t SRV _ldap._tcp.cn.lan
5
```

Lastly KINIT

Create your first user (Optional)

The reason why this is optional, is you have more options when adding a user through the Remote Server Administration Tools (RSAT) on Windows.

```
1 # create a new user in Samba
2 sudo samba-tool user create mkoster
3 # checking users on Samba
4 sudo samba-tool user list
5
```

Windows Setup

PreCheck

1. Set Computer Name
2. Set DNS/IP, The first DNS should be your Samba Server set up above

Verify DNS resolver

```
1 # In Powershell
2 Get-DnsClientServerAddress
3 # ping the AD domain dc1.cn.lan
4 ping dc1.cn.lan
5 # ping the AD domain cn.lan
6 ping cn.lan
7
8
```

Add Server to Directory

After restart login as domain user

Install RSAT to administer domain from Windows

[Download RSAT tools](#) <– For Windows 10 – Could not find a DL source for Windows 11, something I need to look into.

Linux Setup

Precheck

1. On Ubuntu, you may need to add the Universe repositories
 - 1 sudo add-apt-repository universe
 - 2 sudo add-apt-repository multiverse
 - 3 sudo apt update

Set Hostname

```
1 sudo hostnamectl set-hostname fedora.cn.lan
```

Modify Resolved info

On Fedora edit the Resolved config (This works on Ubuntu as well, old editions you may need to change the /etc/resolv.conf file)

```
1 vi /etc/systemd/resolved.conf
2 #add the following
3 [Resolve]
4 DNS=192.168.0.254 9.9.9.9 8.8.8.8
5 sudo systemctl restart systemd-resolved
6
7
```

Install required packages (Ubuntu Only)

For Ubuntu Only

```
1 sudo apt -y install realmd libnss-sss libpam-sss sssd sssd-tools adcli samba-
  common-bin oddjob oddjob-mkhomedir packagekit
```

Discover the Domain (Optional)

```
1 sudo realm discover cn.lan
```

Join the Domain

```
1 sudo realm join -U Administrator cn.lan
```

Restart and you will be able to login as Domain user

```
1 # Default Kerberos Verion 5 Realm:
2 CN.LAN
3 # Kerberos Servers for your realm:
4 dc1.cn.lan
5 # Adminitraive server for your Kerberos realm:
6 dc1.cn.lan
7
8
```



```
1 # rename default Kerberos configuration to krb5.conf.orig
2 sudo mv /etc/krb5.conf /etc/krb5.conf.orig
3 # copy the Kerberos configuration generated by the samba-tool
4 sudo cp /var/lib/samba/private/krb5.conf /etc/krb5.conf
5
```

```
1 # bind the chrony service to IP address of the Samba AD
2 bindcmdaddress 192.168.0.254
3 # allow clients on the network to connect to the Chrony NTP server
4 allow 192.168.0.0/24
5 # specify the ntpsigndsocket directory for the Samba AD
6 ntpsigndsocket /var/lib/samba/ntp_signd
7
8
```

```
1 # restart chronyd service
2 sudo systemctl restart chronyd
3 # verify chronyd service status
4 sudo systemctl status chronyd
5
```

```
1 # authenticate to Kerberos using administrator (MUST BE IN CAPS)
2 kinit administrator@CN.LAN
3 # verify list cached Kerberos tickets
4 klist
5
```

```
1 # add Windows 10 to Active Directory using POWERSHELL
2 Add-Computer -DomainName "cn.lan" -Restart
```